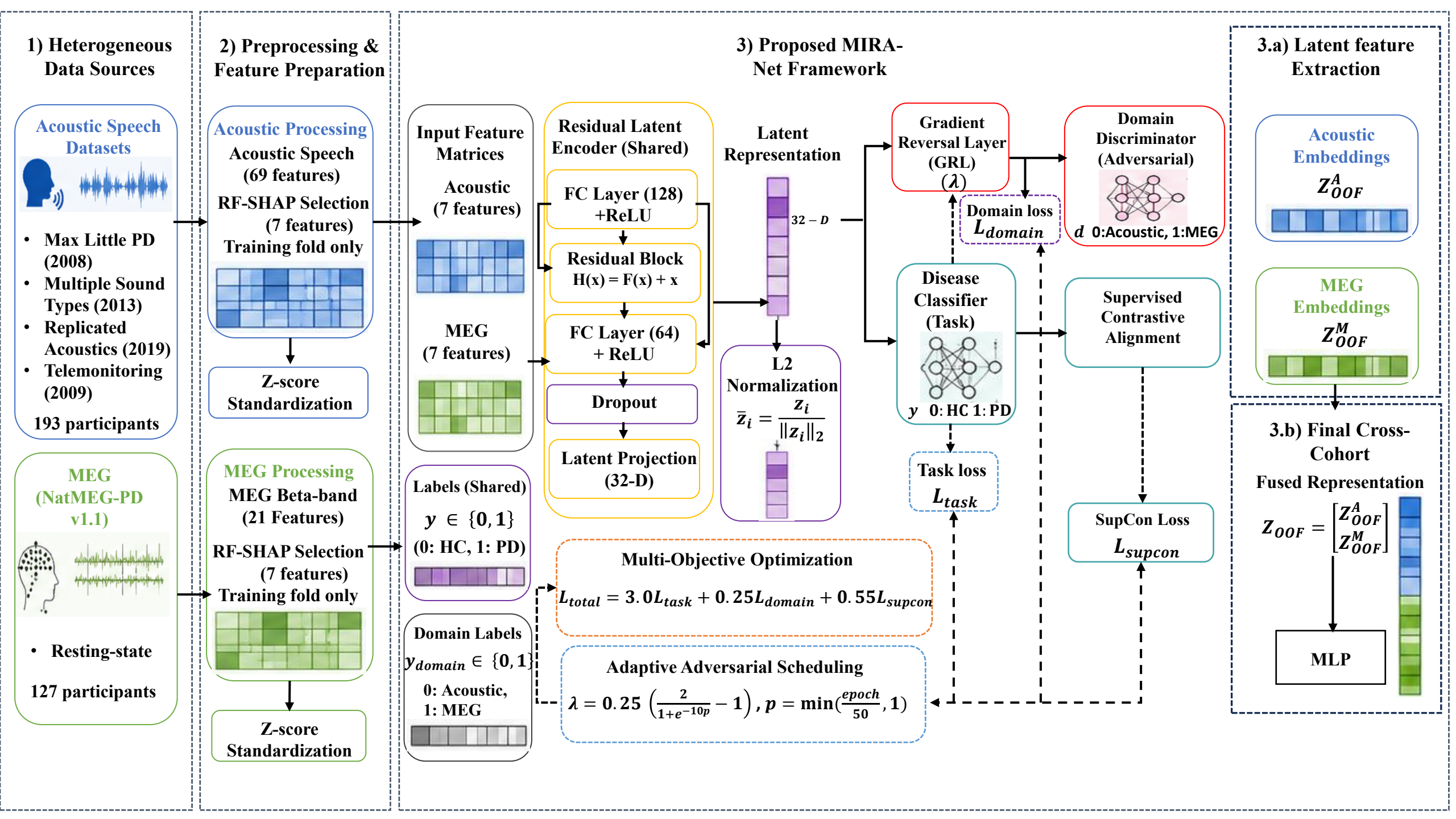

1) Heterogeneous Data Sources
Acoustic Speech Datasets
Max Little PD (2008)
Multiple Sound Types (2013)
Replicated Acoustics (2019)
Telemonitoring (2009)
193 participants
MEG (NatMEG-PD v1.1)
Resting-state
127 participants
2) Preprocessing & Feature Preparation
Acoustic Processing
Acoustic Speech (69 features)
RF-SHAP Selection (7 features)
Training fold only
Z-score Standardization
MEG Processing
MEG Beta-band (21 Features)
RF-SHAP Selection (7 features)
Training fold only
Z-score Standardization
3) Proposed MIRA-Net Framework
Input Feature Matrices
Acoustic (7 features)
MEG (7 features)
Labels (Shared)
$y \in \{0,1\}$
(0: HC, 1: PD)
Domain Labels
$y_{domain} \in \{0,1\}$
0: Acoustic, 1: MEG
Residual Latent Encoder (Shared)
FC Layer (128) +ReLU
Residual Block H(x) = F(x) + x
FC Layer (64) + ReLU
Dropout
Latent Projection (32-D)
Latent Representation
32 – D
L2 Normalization
$\bar{z}_i = \frac{z_i}{\|z_i\|_2}$
Gradient Reversal Layer (GRL) (λ)
Domain loss L_{domain}
Domain Discriminator (Adversarial)
d 0:Acoustic, 1:MEG
Disease Classifier (Task)
y 0: HC 1: PD
Task loss L_{task}
Supervised Contrastive Alignment
SupCon Loss L_{supcon}
Multi-Objective Optimization
$L_{total} = 3.0L_{task} + 0.25L_{domain} + 0.55L_{supcon}$
Adaptive Adversarial Scheduling
$\lambda = 0.25\left(\frac{2}{1+e^{-10p}} - 1\right), p = \min(\frac{epoch}{50}, 1)$
3.a) Latent feature Extraction
Acoustic Embeddings
Z^A_{OOF}
MEG Embeddings
Z^M_{OOF}
3.b) Final Cross-Cohort
Fused Representation
$Z_{OOF} = \begin{bmatrix} Z^A_{OOF} \\ Z^M_{OOF} \end{bmatrix}$
MLP